

QUESTION 1:

Given a binary number, Print its longest non-consecutive digits

Input 1:

101011101010

Output 1:

101010

Input 2:

1010

Output 2:

1010

1111111

10111111000000

0000000111111

QUESTION 2:

Given a matrix of 0's and 1's, that represents an uninfected and infected person respectively in a ward. In a day, an infected person can infect an uninfected person adjacent to it (left, right, top, bottom). Find the minimum number of days required to infect all the people.

Input 1:

0 0 0

0 1 0

0 0 0

Output 1:

2

Explanation:

At day 1,

0 1 0

1 1 1

0 1 0

At day 2,

1 1 1

1 1 1

1 1 1

Total number of days = 2

Input 2:

1 0 0

0 0 0

0 0 0

Output 2:

4

Input 3:

0 0 0

0 0 0

0 0 1

Output 3:

4

Input 4:

1 0 0 0

0 0 0 0

0 0 0 0

0 0 0 1

day 1

1 1 0 0

1 0 0 0

0 0 0 1

0 0 1 1

day 2

1 1 1 0

1 1 0 1

1 0 1 1

0 1 1 1

day 3
1 1 1 1
1 1 1 1
1 1 1 1
1 1 1 1

Output 4:
3

QUESTION 3:

Given a text 'str' and a wildcard pattern 'pattern', your task is to return '1' if the wildcard pattern is matched with text else return '0'. The function takes two strings as arguments 'str' and 'pattern'.

The wildcard pattern can include the characters '?' and '*'

'?' – matches any single character

'*' – Matches any sequence of characters (includes empty sequence). The matching should cover the entire text (not partial text)

Text = "baaabab",
Pattern = "*****ba*****ab", output : 1
Pattern = "baaa?ab", output : 1
Pattern = "ba*a?", output : 1
Pattern = "a*ab", output : 0

QUESTION 4:

The cost of stock on each day is given in an array A[] of size N. Find all the days on which you buy and sell the stock so that in between those days your profit is maximum.

Input:

The first line contains number of test cases T. First line of each test case contains an integer value N denoting the number of days, followed by an array of stock prices of N days.

Output:

For each test case, output all the days with profit in a single line. Also, if there is no profit then print "No Profit".

Constraints:

$1 \leq T \leq 100$
 $2 \leq N \leq 103$
 $0 \leq A_i \leq 104$

Example

Input:

2

7

100 180 260 310 40 535 695

10

23 13 25 29 33 19 34 45 65 67

Output:

(0 3) (4 6)

(1 4) (5 9)

Explanation:

Testcase 1: We can buy stock on day 0, and sell it on 3rd day, which will give us maximum profit.

Note: Output format is as follows - (buy_day sell_day) (buy_day sell_day)

For each input, output should be in a single line.